

Capstone, Presentations & Careers

Module 5 — Putting It Together · Unit 18

The finale. Everything you learned, on one authorized target — then out into the world.

Learning objectives

By the end of this unit, you can:

- **Apply** the full attack lifecycle to an authorized capstone target.
- **Document** your work in a lab journal and a complete pentest report.
- **Present** your findings clearly, framing authorization and impact.
- **Describe** major cybersecurity career paths using CyberSeek and NICE.
- **Sequence** a certification roadmap: Security+ → eJPT/PNPT → OSCP.
- **Identify** competitions, portfolios, and home labs to keep growing.
- **Commit** to staying legal and ethical for life.

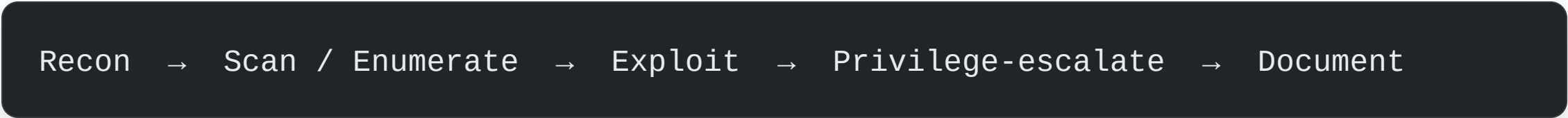
The capstone

Working in **teams of 2-3**, you run the full attack lifecycle on **one authorized target**, then **report** and **present**.

Deliverable	Graded with
Team lab journal (all attempted phases)	Evidence quality
Full pentest report	Report rubric
3-5 min team presentation	Presentation rubric
Careers mini-task + course reflection	(Individual)

A team that captures fewer flags but documents and presents honestly can still earn top marks.

The attack lifecycle



Phase	What you do	Capture
Recon	Read the brief, note in-scope hosts, find the goal	scope, target details
Scan/Enum	<code>nmap</code> , <code>gobuster</code> , manual enumeration	commands + labeled shots
Exploit	Use what you found to gain a foothold; get the flag	steps, payload, flag
Privesc	Escalate — or document <i>honestly</i> that there's none	the path or "none, here's why"

4

Document	Transfer notes into the full report	the complete report
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Capstone targets — approved list ONLY

- **picoCTF** — free, browser-based, year-round practice gym.
- And/or an **instructor-approved beginner TryHackMe room**.

These targets are **legal, isolated, and built to be attacked**.

The approved list is the **entire universe** of legal targets this week. Not sure a target is approved? **Stop and ask**.

CTFs and flags

- **CTF (Capture The Flag):** a legal, gamified security challenge where you solve tasks to find hidden **flags**.
- A **flag** is a token that proves you solved a challenge — often `picoCTF{...}` or `flag{...}`.
- Flags are **proof of completion** — but the **report and presentation** are what's graded.
- **Rules of engagement (RoE):** the agreed boundaries — what you may and may not do.

Write your team scope statement first: *"We are authorized to test only [the approved target]. No other systems are in scope."*

Honest documentation wins

- A team that got **stuck** but wrote up exactly what they tried — with evidence — demonstrates the **real skill**.
- Document **dead ends** and **partial results** truthfully.
- Every screenshot **labeled** (Figure 1, 2, 3...) and tied to the **exact command** that produced it.
- **Reward accuracy over bravado.** Resets from snapshot are free — no penalty.

"Trust me, it worked" is not evidence. Reproducibility is.

The presentation (3-5 min)

A short team talk — build **3-5 slides**:

1. **Target & scope/authorization**
2. **Methodology / lifecycle**
3. **Top findings + severity**
4. **Remediation**
5. **What we learned + ethics**

The **ethics-framing** row is required: explicitly state your **authorization** and the real-world **impact** of your findings.

Cybersecurity career paths

Role	What they do
Pentester / Red team	Simulate attackers to test defenses
SOC analyst / Blue team	Monitor alerts, detect, respond, harden
Purple team	Red + blue working together
GRC	Governance, risk, and compliance — the policy side
AppSec	Find and fix flaws in software/code
IR / Threat intel	Respond to incidents, track adversaries

Explore live demand with **CyberSeek**; map roles with the **NICE Framework**.

The certification roadmap

Security+ → eJPT / PNPT → OSCP → and beyond
(entry-level) (practical pentest) (hands-on pro)

Cert	What it proves
Security+	Broad security fundamentals — a common first step
eJPT / PNPT	Practical, hands-on pentesting at beginner→intermediate
OSCP	The pro version of this course's content — hands-on

Certs open doors, but **a portfolio of real work** proves you can do the job.

Keep growing — for free

- **Competitions:** picoCTF, National Cyber League, CyberPatriot.
- **Home lab:** a safe, isolated practice environment you build yourself.
- **Portfolio:** publish sanitized writeups, lab journals, and projects.
- **College vs. self-study:** both work — **consistency beats either.**

Many great practitioners are self-taught; many went to college. What they share is steady practice and a body of work to show.



Ethics & Authorization — for life

Authorization and scope are the only line between a penetration tester and a criminal.

The Day-1 class rule is now a **life rule**. The same skills, pointed at a real system without **written permission**, are a crime under laws like the **CFAA** — and would end the very career you're aiming for.

Professionals stay employable precisely because they stay in scope, get permission in writing, and disclose responsibly.

Key vocabulary

Term	Meaning
Capstone	The final project — apply everything to one authorized target
Attack lifecycle	Recon → scan/enum → exploit → privesc → document
CTF / Flag	Gamified legal challenge / token proving you solved it
Rules of engagement	The agreed boundaries of a test
Red / Blue / Purple team	Offense / defense / both together
GRC / AppSec	Policy & risk / software security
CyberSeek / NICE	Career map / national catalog of work roles



Lab launch — the capstone

- **Target:** your **picoCTF** set and/or **approved TryHackMe** room — *approved list ONLY.*
- **Read the Safety & authorization reminder aloud** as a team; write your **scope statement**.
- Run the lifecycle: **recon → scan/enum → exploit → privesc → document**.
- Capture **labeled evidence** from minute one; everyone documents.
- Write the **full report** (Unit 17 template) and build your **3-5 slide** deck.

Under deadline pressure, the temptation to "try one more thing" outside scope is real. **Don't.** Scope discipline is what keeps a professional employable.

Recap

- The **capstone** applies the full lifecycle to one **authorized** target.
- **Report + presentation** are the graded deliverables — not flag count.
- **Honest documentation** (including dead ends) is the real skill.
- Careers: **pentest, SOC, red/blue/purple, GRC, AppSec** — explore CyberSeek + NICE.
- Roadmap: **Security+ → eJPT/PNPT → OSCP**; build a **portfolio + home lab**.
- The Day-1 authorization rule is now a **life rule**.

farewell

1. Name the five phases of the attack lifecycle in order.
2. Name one career path and one certification that interest you.
3. What's your **honest next step** after this class?
4. Why does **authorization** protect your future, not just your grade?

You hold powerful skills now. Permission and scope — **in writing, always** — are what keep you on the right side of the line for the rest of your life.

Go build something. Stay curious. Stay legal. You've got this.

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